INCH-POUND
MIL-DTL-24640/14D
22 August 2002
SUPERSEDING
MIL-C-24640/14C
26 June 1995

DETAIL SPECIFICATION SHEET

CABLE, ELECTRIC, 600 VOLTS, LIGHT-WEIGHT, MULTI-CONDUCTOR, AWG 22, TYPE 1XSOW

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of MIL-DTL-24640 listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation.

Construction (Watertight)

First - 7-strand, tin-coated copper conductor. AWG size and dimensions in accordance with Table I.

Second - Insulation material and basic construction requirements shall be in accordance with MIL-W-81044/12, and tested in accordance with the component wire ratings and requirements listed on this slash sheet. Quality conformance testing shall be in accordance with MIL-W-81044, prior to cabling. Component wire color shall be white, unless specified otherwise.

Third - Braided shield of No. 38 AWG tin coated copper strands. Braid coverage shall be no less than 90 percent determined in accordance with WC-27500.

Fourth - Shield insulation of two polyester tapes or a polyester tape and extruded jacket. Identification codes applied using Method 2.

Fifth - The required number of shielded components shall be cabled together. Cabling sequence shall be consecutive, starting with number 1 from center outward. Fillers may be used as necessary, to form a round cable.

Sixth - Binder tape applied helically with overlap.

Seventh - Braided shield, tin coated copper strands. Braid must be constructed to meet Surface transfer impedance requirements as listed herein.

Eighth - Binder tape applied helically with overlap.

Ninth - Cross linked polyolefin jacket. (See Table II for thickness).

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TABLE I. Component wire details.

Watertight conductor			Finished wire			
Wire size (AWG)	Stranding (no. x AWG)	Diameter Nominal (inches)	Diameter (inches)	Test load life cycle accelerated aging tests (lbs. ± 3%)	Low temp. at -40° C	
					Mandrel diameter (inches)	Test load (lbs)
22	7 x 30	0.030	0.047 ± 0.003	0.190	0.75	1.0

TABLE II. Construction details.

PIN M24640/14	Type number	Number of component wires	Diameter over shield (inch) (nom)	Jacket thickness (inch) (min)	Cable diameter (inch)		Cable weight (pounds per 1000 feet) (max)
					(min)	(max)	
-01UO	1XS0W-2	2	0.173	0.034	0.292	0.314	85.0
-02UO	1XSOW-14	14	.362	.041	.470	.506	280.0
-03UO	1XSOW-20	20	.429	.044	.542	.584	330.0
-04UO	1XSOW-30	30	.520	.044	.614	.662	429.0

REQUIREMENTS:

Qualification required.
Construction details: See Tables I and II.

Component wire: The component wire shall be per MIL-W-81044 and shall be tested in accordance with the following wire ratings and additional requirements.

Temperature rating: 135°C (275°F) maximum conductor temperature
Voltage rating: 600 volts (rms) at sea level.
Accelerated aging: Oven temperature, 250 ± 2°C (482 ± 3.6°F) for 6 hours.
Low temp.-cold bend: -40°C (-40°F) for 4 hours
Blocking: 150 ± 2°C (302 ± 3.6°F)
Flammability: 30 seconds (max); 3.0 inches (max); no flaming of tissue paper.
Humidity resistance: 5000 megohms for 1000 feet, minimum insulation resistance after humidity exposure.
Identification, striping or banding durability: 125 cycles (250 strokes) (min), 500 gram weight.
Impulse dielectric test:
Primary insulation (when tested in lieu of spark test):
6.0 kilovolts (peak) 100 percent test.
Finished wire: 8.0 kilovolts (peak), or 3 kHz spark test

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5.5 kV, 100 percent test.
Insulation resistance: 5000 megohms for 1000 feet (min).
Life cycle: Oven temperature, 200 ± 2°C (392 ± 3.6°F) for 120 hours.
Physical properties of primary insulation:
Tensile strength, 2500 lb/in^2 (min).
Elongation, 150 percent (min).
Shrinkage: 0.125 inch maximum at 250 ± 2°C (482 ± 2°F).
Spark test primary insulation: 1500 volts (rms), 60 Hz or 3 kHz, 100 percent test.
Surface resistance: 500 megohms inches (min), initial and final readings.
Thermal shock: Oven temperature, 150 ± 2°C (302 ± 3.6°F).
Maximum change in measurement: 0.060 inch.
Wet dielectric test: 2500 volts (rms).
Wrap test: Mandrel wrap test required.

EXAMINATION AND TESTS:

Basic electrical:	Requirements:
Conductor Resistance – ohms/1000 feet at 25°C, maximum	17.63
Voltage withstand - volts, root mean square, minimum	
Conductor to conductor	Not applicable
Conductor to shield	1000
Component Shield to Component Shield	200
Component Shield to overall shield	500
Insulation resistance – megohms-1000 feet, minimum	
Conductor to shield	2500
Shield to shield (component shield only)	100
Shield to shield (component shield to overall shield)	100
Conductor and shield continuity	No failure
Jacket flaws	No failure
Group A:	
Visual and dimensional	No failure
Capacitance:	
Mutual capacitance - at 1 MHz, picofarads/foot (pF/ft), maximum	120
Capacitance unbalance - percent, maximum	Not applicable
Characteristic impedance - at 1 MHz, ohms	13 min, 27 max
Watertightness (see Table IX of MIL-DTL-24640 for limits of water leakage)	No failure
Weight	No failure

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Group B:

- Cold bending ..... No failure
- Drip ..... No failure
- Cross link proof test (percent maximum)
  - Jacket 50
- Tear - pounds per inch thickness, minimum 35
- Physicals (unaged)
  - Jacket
    - Tensile strength - lb/in^2, minimum 1300
    - Elongation - percent, minimum 160
- Attenuation - at 3 MHz, dB/100 feet, maximum ..... Not applicable
- Gas flame (1 hour) ..... Not applicable

Group C:

- Physicals (aged)
  - Jacket
    - Air oven
      - Tensile strength - percent of unaged, minimum 60
      - Elongation - percent of unaged, minimum 60
- Permanence of printing (conductor) - cycles, minimum 25
- Permanence of printing (jacket) - cycles, minimum 125
- Shield - conformance to material, construction and coverage ..... No failure
- Heat distortion - percent of unaged, maximum 30
- Cable sealant removability ..... No failure
- Shrinkage ..... No failure
- Mutual inductance - dB/1000 feet, maximum ..... Not applicable
- Surface transfer impedance
  - Milliohms per meter, maximum 700
  - EMP response, dB, minimum 60
- Pulse response time - ms/500 feet
  - Rise time, maximum ..... Not applicable
  - Decay time, maximum ..... Not applicable

QUALIFICATION INSPECTION:

Qualification inspection shall include basic electrical, all of groups A, B, C and D, plus the following:

- Flame propagation ..... No failure
- Aging and compatibility (cable) ..... No failure
- Acid gas equivalent - percent, maximum
  - Jacket 2
  - Fillers 2
  - Insulation 18
- Halogen content - percent, maximum

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Jacket	0.2
Fillers	0.2
Hot oil immersion	
Tensile strength - percent of unaged, minimum...	50
Elongation - percent of unaged, minimum	50
Smoke index, maximum	
Jacket	25
Fillers	45
Insulation	10
Toxicity index, maximum	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) - (type A) hardness, minimum	80
Weathering (jacket)	No failure
Fungus Resistance	No failure
Abrasion resistance	No failure
Gas Flame (3 hour)	Not applicable
Component wire	No failure

Unit ordering length: 500 feet nominal.

Custodians:
Army - MI
Navy - SH

Preparing Activity:
Navy - SH
(Project 6145-2307-014)

Review Activities:
Army - AR, AV, CR, CR4
Navy - CG, EC
DLA - IS

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